MARIAPPAN V

Senior Data Quality Engineer | ETL & Data Warehouse Validation | PySpark & Python Automation

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PROFESSIONAL SUMMARY

Accomplished **Data Quality Engineer** with 5+ years of specialized engineering experience in data warehouse testing, ETL/ELT pipeline validation, and robust test automation frameworks across distributed cloud ecosystems. Expert in designing automated validation engines leveraging **Pytest, Python, PySpark, and advanced SQL** on **AWS** and **Databricks**. Proven track record of reconciling large-scale datasets across relational stores, NoSQL (MongoDB), and object stores (Parquet, JSON, CSV), cutting validation cycle times by **60%** and maintaining **99.99% data integrity**. Leverages **GitHub Actions** for CI/CD test automation, **Git/GitHub** for version control, and **GitHub Copilot** agent for AI-accelerated test script development and SQL optimization in fast-paced Agile environments.

TECHNICAL EXPERTISE

Data Quality & Testing:	ETL & Data Warehouse Testing, Data Parity Reconciliation, Schema Drift & Evolution Validation, Nullability & Constraint Testing, Pytest Test Automation, Data Profiling, Anomaly Detection
Programming & AI:	Python, PySpark, Advanced SQL (Window Functions, CTEs, Execution Plans), GitHub Copilot (Agent Coding & Prompt Engineering), DuckDB, HQL
Cloud & Big Data:	AWS (S3, Glue, Lambda, Athena), Databricks (Delta Lake, Workflows, Notebooks), Azure Data Factory, Hadoop Ecosystem (HDFS, Hive, Spark)
CI/CD & Versioning:	GitHub (Git Versioning, Code Review, Branching Strategy), GitHub Actions (Automated CI/CD Test Pipelines), JIRA, Agile/Scrum
Databases & Formats:	MongoDB, PostgreSQL, MySQL, Oracle, Delta Lake, Apache Parquet, JSON, CSV, Excel
BI & Automation:	Power BI, Advanced Excel Modeling, Power Automate, SharePoint, Linux/Shell Scripting

PROFESSIONAL EXPERIENCE

Development Engineer (Data Quality & Test Automation)

2024 – Present

Comcast India Engineering Center | Chennai, India

- Architected and automated end-to-end data validation frameworks using **Pytest and Python**, slashing regression testing cycle times by **60%** while guaranteeing **99.99% data integrity** across enterprise data lakes and warehouses.
- Engineered distributed ETL/ELT validation pipelines leveraging **PySpark, Python, and SQL** on AWS and Hadoop clusters, validating and reconciling multi-million record batches daily across raw, cleansed, and curated reporting layers.
- Established automated Continuous Integration (CI) testing by implementing **GitHub Actions** workflows, triggering automated Pytest validation suites and reconciliation checks on pull requests and pipeline runs.
- Accelerated test automation coverage, script generation, and complex SQL validation queries by leveraging **GitHub Copilot Agent**, increasing test framework development velocity.
- Managed multi-branch source code repositories, automated pull-request validation gates, and versioning standards utilizing **Git and GitHub**.
- Built comprehensive cross-system reconciliation checks across heterogeneous environments, validating business transformations and parity across **MongoDB, SQL databases (PostgreSQL/Oracle), Parquet, CSV, and Excel**.
- Conducted root cause analysis on data anomalies (schema drift, orphan records, missing keys, boundary constraint violations) in Agile sprints via **JIRA**, resolving over **200+ data quality bottlenecks**.
- Standardized enterprise Data Quality (DQ) metrics, automated testing guidelines, and validation documentation for downstream analytics stakeholders.

PROFESSIONAL EXPERIENCE (CONTINUED)

ML Analyst (Data Engineering & Validation)
2021 – 2024

Likewize | Chennai, India

- Leveraged **PySpark, Hive, and HDFS** for distributed data processing, automated data profiling, and quality validation across high-volume structured and semi-structured datasets.
- Developed and executed complex **SQL validation scripts** to audit multi-table warehouse joins, transformations, aggregations, and business metrics, tuning queries to reduce execution runtimes by **40%**.
- Automated end-to-end data ingestion and reconciliation workflows using **Python ETL scripts** across diverse file formats including JSON, Parquet, and CSV, mitigating pipeline ingestion failures.
- Designed interactive **Power BI dashboards** and SQL audit reports delivered to 50+ enterprise stakeholders to track data quality SLAs, data completeness, and trend variances.
- Collaborated closely with data engineering and business teams to audit feature pipelines, ensuring data consistency and reliability for machine learning models and reporting systems.

Data Analyst
2019 – 2021

Capgemini Business Services | Chennai, India

- Engineered automated operational reporting and data verification workflows using **Power BI and Advanced Excel**, boosting operational reporting efficiency and audit readiness by **35%**.
- Constructed automated notification and audit workflows using **Power Automate** and SharePoint to systematically validate, track, and distribute operational reports.
- Formulated KPI tracking mechanisms and automated exception detection rules to identify business anomalies and data inconsistencies prior to management reviews.
- Conducted user acceptance testing (UAT) and reconciliations between source operational databases and reporting extracts.

Senior MIS Associate
2014 – 2019

Tata Consultancy Services (TCS) | Chennai, India

- Managed operational data reconciliation, KYC compliance audits, and business process performance reporting for capital markets and banking operations.
- Built robust Excel macro solutions (VBA) and automated data auditing templates that streamlined daily operational validation and eliminated manual reconciliation errors.
- Delivered daily and monthly SLA performance metrics, attendance tracking models, and variance reports for executive leadership.

EDUCATION

Bachelor of Commerce (Professional Accounting)
K.R. College of Arts & Science, Manonmaniam Sundaranar University | 2011 – 2014

Declaration: I hereby declare that all the information furnished above is true, complete, and correct to the best of my knowledge and belief. — *Mariappan V*